Biochemistry Cumulative Examination

September 29, 2006

On Monday, September 18, 2006, George Makhatadze from Hershey Medical Center presented a seminar. Answer the following questions based upon your general biochemical knowledge or upon his presentation.

1. Dr. Makhatadze presented a method for increasing the stability of proteins. Briefly, why is increasing protein stability of interest to anyone? List two specific potential applications. 20 points.

2. How did Dr. Makhatadze increase protein stability? Specifically, what kind of interactions did he focus on? 15 points.

3. Dr. Makhatadze focused on making stabilizing mutations on the surface of the protein. According to him, what is/are the advantage(s) of working in this area of the protein, as opposed to the protein core? 20 points

4. Dr. Makhatadze used a very simple computational technique to determine which amino acids to mutate. As he acknowledged, there are much more sophisticated modeling programs available out there which are much more realistic. Why did he use the more primitive calculation method in his work? 10 points.

5. Dr. Makhatadze frequently used a short hand to describe the various mutants they made and studied, for example, K122E. Explain what each part of this notation means. 15 points.

6. Dr. Makhatadze described several mutants that had gained quite a bit of stability. To the nearest order of magnitude, how many kJ/mol of stabilization did they achieve? To the nearest order of magnitude what is the usual stability (difference in the free energy of the native and denatured states) of a protein from a mesophilic organism? 10 points.

7. Dr. Makhatadze talked about mesophiles and thermophiles quite a bit. Briefly define these terms. 10 points.